

Code:

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#include <stdio.h> #include <stdlib.h>

struct node { int data; struct node *next; };

void create(struct node **head, int data) { struct node new_node = (struct
node)malloc(sizeof(struct node)); new_node->data = data; new_node->next = NULL; if
(*head == NULL) { *head = new_node; } else { struct node *temp = *head; while (temp-
>next != NULL) { temp = temp->next; } temp->next = new_node; } }

void print(struct node *head) { struct node *temp = head; while (temp != NULL)
{ printf("%d", temp->data); if (temp->next != NULL) printf(" "); temp = temp->next; }
printf("\n"); }

struct node* reverseList(struct node* head) { struct node *prev = NULL, *curr = head,
*nxt = NULL; while (curr != NULL) { nxt = curr->next; curr->next = prev; prev = curr; curr =
nxt; } return prev; }

void fold(struct node **head) { if (*head == NULL || (*head)->next == NULL) return;

struct node *slow = *head;
struct node *fast = *head;
struct node *prev_slow = NULL;

while (fast != NULL && fast->next != NULL) {
    fast = fast->next->next;
    prev_slow = slow;
    slow = slow->next;
}

// Split into two halves
if (fast != NULL) {
    prev_slow = slow;
    slow = slow->next;
}
prev_slow->next = NULL;

struct node *second = reverseList(slow);
struct node *first = *head;

// Alternate merge
while (second != NULL) {
```

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    struct node *t1 = first->next;
    struct node *t2 = second->next;

    first->next = second;
    if (t1 == NULL) break;
    second->next = t1;

    first = t1;
    second = t2;
}

}

int main() { int n; if (scanf("%d", &n) != 1) return 0;

struct node *head = NULL;
for (int i = 0; i < n; i++) {
    int val;
    scanf("%d", &val);
    create(&head, val);
}

printf("Link list data:");
print(head);

fold(&head);

printf("Link list data after fold:");
print(head);

return 0;

}

```

Input:

16

1 3 4 7 9 11 13 14 18 19 23 24 56 32 98 17

Output:

Link list data:1 3 4 7 9 11 13 14 18 19 23 24 56 32 98 17

Link list data after fold:1 17 3 98 4 32 7 56 9 24 11 23 13 19 14 18